

Network Performance Optimization

Operational Delay Analysis & Network Performance Improvement — a data-driven review of route, hub, and time-based delay patterns across the 2015 operating year.

PREPARED BY

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DATA SOURCE

U.S. DOT On-Time Performance

2015 Operating Year



REPORT

Table of Contents

18 sections covering the full analytical workflow from data ingestion to business recommendations.

01	Executive Summary Key findings & methodology	10	Route Performance Highest delay-rate routes
02	Business Problem Operational reliability imperatives	11	Airport Delay Analysis Delay rate by departure airport
03	Project Workflow Ten-stage analytical process	12	Time-Based Delay Delay rate by departure hour
04	Dataset Overview Scope, source & structure	13	Monthly Delay Trends Seasonality across 2015
05	Data Preparation Cleaning & standardization	14	Key Insights Six operational findings
06	Feature Engineering Seven analytical features	15	Business Recommendations Prioritized action plan
07	Operational Performance Baseline KPIs: 2015	16	Business Impact Expected qualitative benefits
08	Busiest Routes Top 10 by flight volume	17	Conclusion Summary & implications
09	Hub Airport Analysis Departure & arrival volumes	18	Appendix Methodology & data notes

Executive Summary

United Airlines operates one of the largest domestic route networks in the country, connecting passengers through five primary hubs — Chicago O'Hare, Houston Intercontinental, Denver, San Francisco, and Newark Liberty.

This engagement analyzes **472,355 United Airlines flights** operated in 2015, drawn from the U.S. Department of Transportation's On-Time Performance dataset, to identify where delays originate, how they propagate through the network, and which routes, airports, and time windows carry the greatest operational risk.

OBJECTIVE

Provide network planning and analytics leadership with a data-driven view of operational reliability — sufficient to prioritize where schedule, staffing, and turnaround investment will have the greatest network-wide effect.

21.10%

Delay Rate

Nearly 1 in 5 flights arrived 15+ minutes late in 2015.

1.37%

Cancellation Rate

Cancellations were rare, indicating broadly stable operations.

ORD

Core Network Bottleneck

~59,500 departures — the largest concentration of network risk.

6–8 PM

Evening Peak Risk

Delay rates climb from ~8% at 5 AM to ~29–30% in the evening.

The Business Problem

Flight delays are not simply a customer experience issue — they carry direct and indirect operational costs across the business.

Customer Dissatisfaction Delays erode passenger trust and satisfaction scores, particularly on high-frequency business routes.	Missed Connections Hub-and-spoke operations mean a single delay can cascade into missed connections across the network.	Increased Crew Costs Delays trigger crew rescheduling, overtime, and duty-time limitations that raise operating costs.
Aircraft Utilization Issues Late arrivals compress turnaround windows, reducing the number of productive flight hours per aircraft.	Network Disruptions Because aircraft and crews rotate across multiple legs, localized delays propagate system-wide.	Higher Operating Costs Fuel burn, gate fees, and re-accommodation costs all rise as delays accumulate through the day.

PROJECT OBJECTIVE

Analyze operational performance across United's network to identify primary delay drivers, measure key KPIs, evaluate disruption concentration by route, airport, and time window, and translate findings into actionable, data-driven operational recommendations.

Project Workflow

A ten-stage analytical process from problem definition through to business recommendations.

- 01

Business Problem
Define the operational reliability challenge and analytical objectives.
- 02

Data Collection
Source 2015 DOT On-Time Performance and airport reference datasets.
- 03

Data Cleaning
Filter to United Airlines flights and resolve data quality issues.
- 04

Feature Engineering
Construct route, delay flag, cancellation flag, time-of-day, and distance features.
- 05

Exploratory Analysis
Establish baseline KPIs for delay rate, cancellation rate, and arrival delay.
- 06

Route Performance Analysis
Identify the busiest routes and those with the highest delay rates.
- 07

Airport Performance Analysis
Evaluate hub-level departure volume and delay concentration.
- 08

Time-Based Analysis
Examine how delay rates vary by hour of day and by month.
- 09

Business Recommendations
Translate findings into prioritized operational actions.
- 10

Conclusion
Summarize opportunities for network-wide reliability improvement.

Dataset Overview



Source

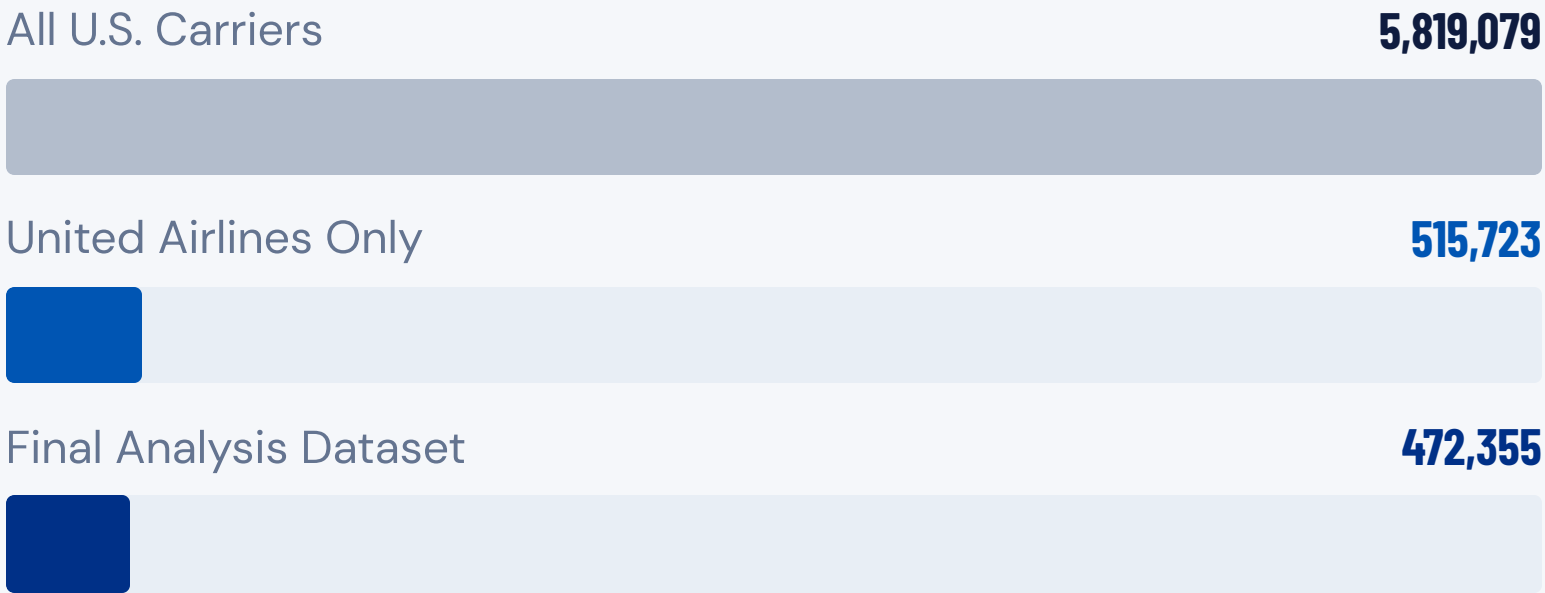
The U.S. Department of Transportation's On-Time Performance dataset for 2015, covering more than 5.8 million flights across all U.S. carriers, paired with a 322-airport reference table.

Data Refinement

Filtering to United's carrier code ("UA") isolates 515,723 flights. Approximately 43,000 October 2015 records using numeric DOT airport IDs — rather than standard IATA codes — were excluded to ensure route-level analysis reliability.

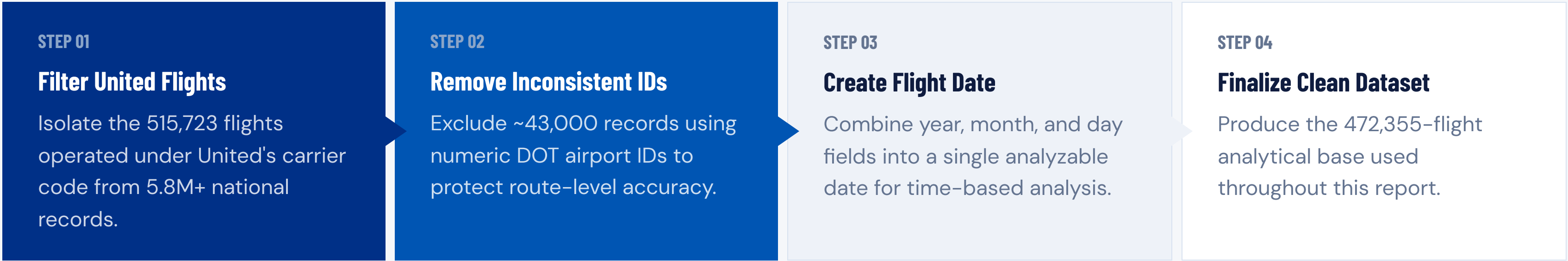
Final Dataset
472,355 flights — sufficient to support robust route, hub, and time-based analysis across the full 2015 operating year.

Data Narrowing



Bar widths are illustrative of relative scale, not linear.

Data Preparation



Before analysis, the raw DOT records required systematic preparation. Flights were first filtered to United Airlines' operations, then reviewed for data quality issues. A unified flight date was constructed from the year, month, and day fields to support time-based analysis.

Several delay-cause fields — including late-aircraft, weather, airline, air-system, and security delay — are populated only when a flight is delayed or cancelled, and were reviewed accordingly. Core scheduling fields were essentially complete, providing a reliable foundation for subsequent KPI calculations.

DATA QUALITY NOTE

Approximately 43,000 October 2015 records used numeric DOT Airport IDs instead of IATA codes. These were excluded rather than reconciled, prioritizing consistency and reliability over completeness.

Feature Engineering

Seven engineered features underpin every analysis in this report.

FEATURE	PURPOSE	BUSINESS VALUE
Flight Date	Unifies year, month, and day into a single date field.	Enables monthly and seasonal trend analysis across the operating year.
Route	Concatenates origin and destination into an origin — destination pair.	Supports route-level performance ranking and prioritization.
Delay Flag	Flags flights arriving 15+ minutes late (99,680 of 472,355 flights).	Provides the industry-standard definition of a "delayed" flight for KPI reporting.
Cancellation Flag	Flags cancelled flights (6,465 of 472,355 flights).	Separates cancellation risk from delay risk in operational reporting.
Departure Hour	Extracts the scheduled departure hour from the schedule field.	Enables hour-by-hour delay pattern analysis across the operating day.
Time of Day	Buckets departures into Morning, Afternoon, Evening, and Night.	Simplifies scheduling discussions for non-technical stakeholders.
Distance Category	Classifies flights as Short, Medium, or Long Haul by distance.	Allows delay drivers to be compared across stage-length.
DISTRIBUTION NOTE	By time-of-day: Morning 42.4% (200,175), Afternoon 28.5% (134,569), Evening 21.7% (102,719), Night 7.4% (34,892). By distance: Medium-Haul 52.1% (245,993), Long-Haul 34.1% (160,903), Short-Haul 13.9% (65,459).	

Operational Performance Overview



United maintained a low cancellation rate of 1.37% — indicating broadly stable, predictable operations at the network level. Delay performance tells a different story: **21.10% of flights, roughly 1 in every 5, arrived 15 minutes or more behind schedule.**

The average arrival delay of 6.20 minutes is modest, but heavily influenced by the four in five flights that arrived on time. Among delayed flights, the operational and customer impact is materially larger — and concentrated in specific hubs, routes, and time windows rather than spread evenly across the network.

BUSINESS INSIGHT

A sub-2% cancellation rate suggests United's core scheduling and fleet reliability are sound; the operational challenge is concentrated in on-time performance rather than flight completion.

RECOMMENDATION

Prioritize delay-reduction initiatives over cancellation-reduction initiatives, and target them at the specific hubs, routes, and hours identified in subsequent sections.

Busiest Routes

Top 10 Busiest United Airlines Routes – Flight Volume 2015



BUSINESS INSIGHT

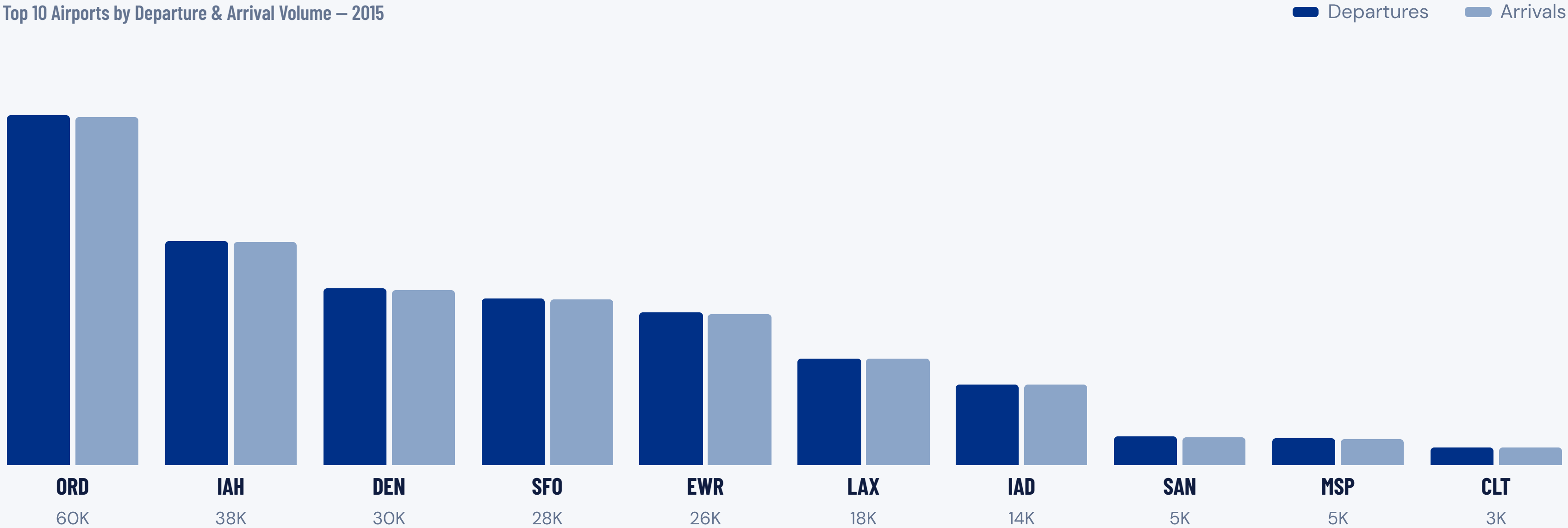
The busiest routes cluster around four hubs — ORD, SFO, IAH, and EWR — with SFO–ORD and ORD–SFO alone accounting for nearly 8,800 flights in 2015.

RECOMMENDATION

Treat the top 10 routes as the primary lever for network-wide reliability gains. Turnaround, gate, and crew-planning improvements should be piloted here first.

Hub Airport Analysis

Top 10 Airports by Departure & Arrival Volume – 2015



BUSINESS INSIGHT

ORD, IAH, DEN, SFO, and EWR anchor the network. Departure and arrival volumes are closely mirrored at every airport, confirming a classic hub-and-spoke structure.

RECOMMENDATION

Hub coordination should be managed as a network-level function. Investments in ORD, IAH, and DEN turnaround reliability will have outsized network effects given their combined departure share.

Route Performance Analysis

Highest delay-rate routes among those with 100 or more flights — 2015

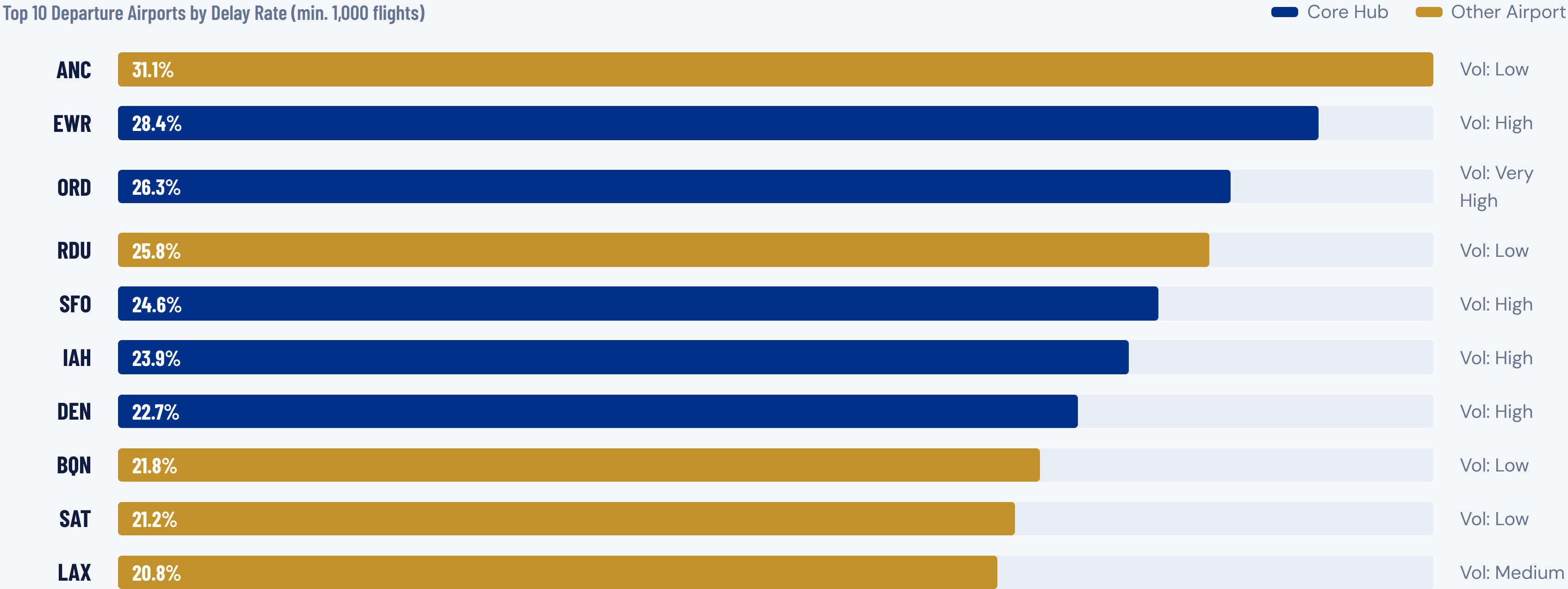
ROUTE	FLIGHTS	DELAY RATE	AVG. ARRIVAL DELAY	CANCELLATION RATE	
ORD — ANC	345	41.74%	16.61 min	0.58%	HIGH
ORD — SAT	208	40.87%	20.67 min	0.96%	HIGH
RDU — EWR	182	40.66%	37.56 min	1.65%	HIGH
EWR — BQN	291	39.86%	19.16 min	1.37%	HIGH
ANC — ORD	346	39.02%	18.23 min	1.45%	HIGH

The highest delay rates appear on longer-haul or lower-frequency segments — ORD–ANC, ORD–SAT, and RDU–EWR each exceeding a 39% delay rate, roughly double the network average. The RDU–EWR pairing stands out with a 37.56-minute average arrival delay, suggesting when this route is late, it tends to be very late.

INTERPRETATION

These patterns may indicate a combination of airspace congestion, seasonal weather exposure, and tighter scheduling buffers on lower-frequency city pairs. Validate against airline-controllable delay-reason data before schedule changes are made.

Airport Delay Analysis



BUSINESS INSIGHT

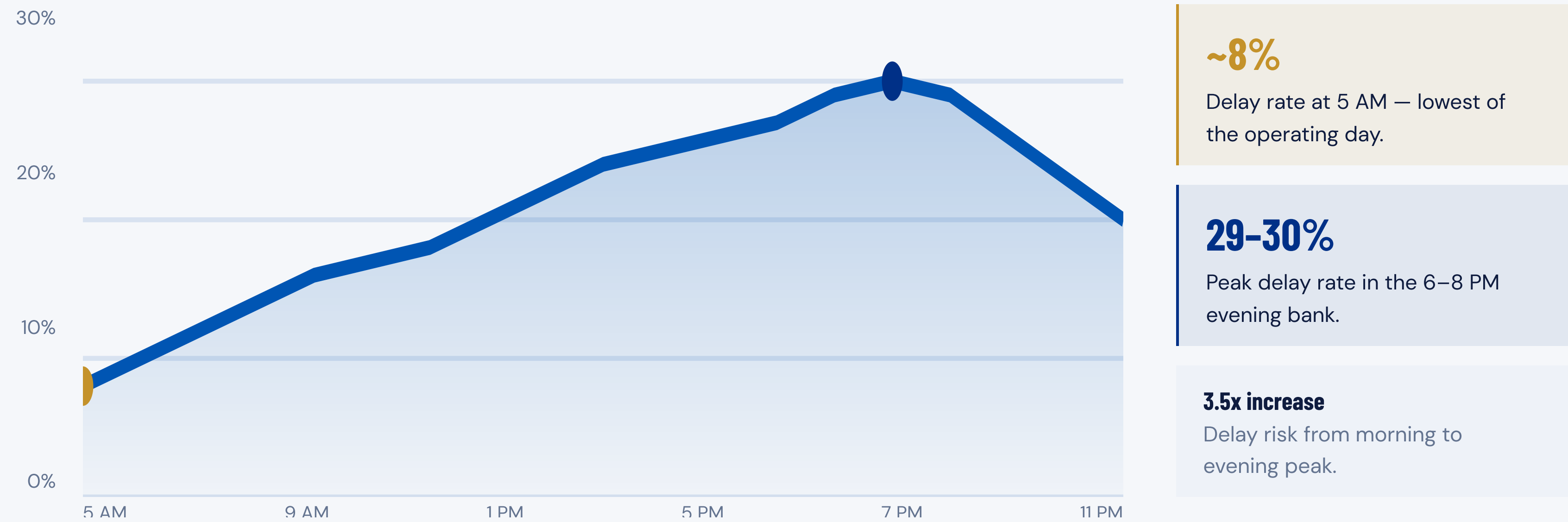
ANC has the highest rate at 31.1%, but ORD combines a 26.3% rate with ~59,500 departures — the largest volume in the network. Delay rate alone is a misleading prioritization signal.

RECOMMENDATION

Prioritize operational investment by delay volume (rate × traffic), not rate alone. On this basis: ORD first, followed by IAH, EWR, and DEN.

Time-Based Delay Analysis

Delay Rate by Scheduled Departure Hour



INTERPRETATION

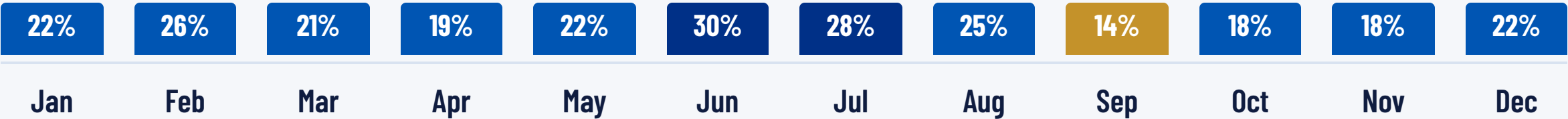
Consistent with delay propagation: aircraft and crews that start on schedule tend to stay on schedule; any disruption compounds across subsequent legs, gate turns, and crew connections.

RECOMMENDATION

Focus recovery actions earlier in the day. Consider added schedule buffer and resource flexibility specifically in the late-afternoon and evening banks.

Monthly Delay Trends

Monthly Delay Rate — 2015 (% flights arriving 15+ min late)



PEAK MONTH

June

~30% delay rate

TROUGH MONTH

September

~14% delay rate

2.1x variation

Peak-to-trough ratio across the operating year.

INTERPRETATION

The June–July peak is consistent with peak summer demand and seasonal thunderstorm activity. September's trough may reflect the post-summer travel lull. These are correlations, not confirmed causal drivers.

RECOMMENDATION

Build seasonal staffing and schedule-buffer planning around the June–July risk window. Validate the weather hypothesis against the dataset's weather-delay field before finalizing changes.

Key Insights

The six findings that most shape United's operational priorities.

Critical Hubs

ORD, IAH, DEN, SFO, and EWR anchor the network and carry the greatest concentration of operational risk.

Highest-Risk Routes

ORD–ANC, ORD–SAT, RDU–EWR, EWR–BQN, and ANC–ORD each exceed a 39% delay rate — roughly double the network average.

Peak Delay Hours

Delay risk is structurally lowest in early morning (~8%) and highest in the 6–8 PM evening bank (~29–30%).

Delay Propagation

Delays compound through the day as aircraft and crews carry disruption forward across legs — a structural characteristic of the hub-and-spoke model.

Network Bottleneck

ORD combines the network's highest departure volume (~59,500) with an above-average delay rate of 26.3% — the single largest source of network risk.

Seasonal Peak

June delay rates (~30%) are more than double the September low (~14%), representing a 2.1x peak-to-trough variation across the operating year.

Business Recommendations

RECOMMENDATION	BUSINESS RATIONALE	EXPECTED BENEFIT	PRIORITY
Improve turnaround efficiency at ORD	ORD has the highest departure volume (~59,500) with a 26.3% delay rate.	Largest network-wide reduction in propagated delay per unit of investment.	HIGH
Add schedule buffer on the 6-8 PM evening bank	Delay rate climbs from ~8% at 5 AM to ~29-30% between 6-8 PM.	Reduced evening delay severity and fewer downstream misconnections.	HIGH
Strengthen aircraft rotation planning	Delay propagation is consistent with disruption carried forward across legs.	Fewer late-aircraft delays inherited from earlier rotations.	HIGH
Deepen hub coordination across ORD, IAH, DEN, SFO, EWR	These five hubs anchor the large majority of departures and connecting itineraries.	More resilient network recovery during localized disruptions.	HIGH
Apply targeted buffers to the five highest-risk routes	ORD-ANC, ORD-SAT, RDU-EWR, EWR-BQN, and ANC-ORD each exceed a 39% delay rate.	Improved reliability on the routes most likely to disrupt connecting itineraries.	MED
Build a predictive delay-risk dashboard	Route, hour, month, and distance features already show strong, distinct delay signals.	Earlier operational awareness and more proactive schedule recovery.	MED
Plan seasonal staffing around the June-July risk window	Monthly delay rate peaks near 30% in June versus ~14% in September.	Better-matched resourcing ahead of the highest-risk travel season.	MED
Monitor high-delay thin routes with periodic review	Several lower-frequency routes show delay rates roughly double the network average.	Early detection of emerging reliability issues on smaller city pairs.	LOW

Business Impact

Qualitative benefits expected from sustained execution of these recommendations.

Higher Operational Reliability

A more predictable network with fewer disruptions cascading from peak hubs and hours.

Better Customer Satisfaction

Fewer missed connections and delayed arrivals on the routes passengers fly most.

Lower Propagated Delays

Reduced carry-forward disruption across aircraft rotations and connecting itineraries.

Improved Aircraft Utilization

More productive flight hours per aircraft as turnaround reliability improves.

Reduced Crew Disruptions

Fewer schedule breaks, less overtime exposure, and more predictable crew planning.

Improved On-Time Performance

A measurable reduction in the 21.10% baseline delay rate over time.

LOWER OPERATIONAL COSTS

Reduced re-accommodation, gate, and irregular-operations costs tied to delay recovery across the entire network.

Impact described qualitatively.
This analysis does not estimate financial savings.

Conclusion

This analysis examined 472,355 United Airlines flights operated in 2015 to understand where, when, and how operational delays concentrate across the network. Rather than treating delay as a uniform phenomenon, the data shows that risk is **concentrated** — in specific hubs, on specific routes, during specific hours, and across specific months.

The analysis successfully identified:

- **Operational bottlenecks** — most notably at Chicago O'Hare, where high departure volume and an elevated delay rate combine to create the network's single largest source of risk.
- **Critical hubs** — ORD, IAH, DEN, SFO, and EWR, which together anchor the large majority of crew rotations, aircraft rotations, and connecting itineraries.
- **Time-based delay propagation** — a clear intraday pattern showing how early disruptions compound into evening-peak delay rates of 29–30%.
- **Seasonal concentration** — a June–July risk window with delay rates more than double the September low, enabling proactive seasonal resource planning.

EXPECTED VALUE

Applied consistently, the eight recommendations are intended to reduce delay propagation, strengthen on-time performance, and improve aircraft and crew utilization across United's core hub network.

Data Source

U.S. Department of Transportation On-Time Performance dataset, 2015. Final analytical base: 472,355 flights across 322 airports.

Caveat

Findings reflect correlational patterns in the 2015 dataset. Causal validation requires additional operational data before final schedule or staffing changes are made.

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U.S. DOT On-Time Performance · 2015 Operating Year

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ANALYTICAL SCOPE

472,355 Flights · 322 Airports

January – December 2015

CLASSIFICATION

Appendix

Slide 20 of 20